

BCA SEMESTER V • INTRODUCTION TO MACHINE LEARNING • UNIT IV

Discrete Features, Multivariate Regression and Dimensionality Reduction

Comprehensive classroom lectures aligned strictly with prescribed textbook curriculum.



Primary Reference: **Ethem Alpaydin**, *Introduction to Machine Learning* (2004), PHI / MIT Press

Unit IV Learning Roadmap



PART A • 5.7

Discrete Features

Bernoulli models and naive conditional independence.



PART A • 5.8

Multivariate Regression

Multiple continuous responses and least-squares weights.



PART B • 6.1

Dimensionality Intro

Selection versus extraction core paradigms.



PART B • 6.2

Subset Selection

Forward and backward greedy search algorithms.



PART B • 6.3

PCA Analysis

Eigenvector projection and variance maximization.

Continuous vs Discrete Features

The Modeling Challenge

Continuous features assume real-valued inputs modeled by multivariate Gaussian densities.

Discrete features represent categorical states and require distinct probability mass formulations.

Property	Continuous Features	Discrete Features
Domain	Real numbers	Finite states
Density Model	Multivariate Normal	Bernoulli or Multinomial
Parameters	Mean and Covariance	Class conditional probabilities

Textbook Modeling of Discrete Features

1. Bernoulli Representation

Binary feature vector elements take values 0 or 1.

$$P(x_j | C_i) = p_{ij}^{x_j} (1 - p_{ij})^{(1 - x_j)}$$

2. Naive Bayes Assumption

Features are conditionally independent given the class.

$$p(x | C_i) = \prod_{j=1}^d p(x_j | C_i)$$

3. Parameter Estimation

Parameters estimated directly using sample proportions.

$$\hat{p}_{ij} = \sum_t x_{jt} r_{it} / \sum_t r_{it}$$

Estimation for Discrete Features

Training Samples: Class C1 (Loan Approved)

Sample	x1 (Employed)	x2 (Owns House)
t = 1	1	1
t = 2	1	0
t = 3	1	1
t = 4	0	1

Step-by-Step Probability Calculation

1. **Parameter Estimates:** Total class count $N_1 = 4$

$\hat{p}_{11} = 3 / 4 = 0.75 \mid \hat{p}_{12} = 3 / 4 = 0.75$

2. **New Applicant Vector:** $x = [1, 0]^T$

$P(x \mid C_1) = (0.75)^1(0.25)^0 \times (0.75)^0(0.25)^1$

Result: $P(x \mid C_1) = 0.75 \times 0.25 = 0.1875$

Multiple vs Multivariate Regression

Unit III Concept

Multiple Linear Regression

Predicts a single scalar continuous target r from input vector x .

$$r = W^T x + \epsilon$$

Section 5.8 Formulation

Multivariate Regression

Predicts multiple continuous target outputs $r = [r_1, \dots, r_k]^T$ simultaneously.

$$r = W^T x + \epsilon$$

The Multivariate Linear Model

Matrix Formulation

For sample t , target outputs are modeled as linear combinations:

$$r_{jt} = \mathbf{w}_j^T \mathbf{x}_t + \epsilon_{jt}$$

In full matrix form over all N training instances:

$$\mathbf{R} = \mathbf{XW} + \mathbf{E}$$

Symbol Notation

X: $N \times (d+1)$ input data matrix with bias column

W: $(d+1) \times k$ parameter matrix of linear weights

R: $N \times k$ response target matrix

E: $N \times k$ zero-mean residual noise matrix

Least Squares Parameter Estimation

1. Error Criterion

Minimize sum of squared errors across all outputs.

$$E(W) = \frac{1}{2} \sum_t \| r_t - W^T x_t \|^2$$

2. Normal Equations

Take derivative with respect to W and set to zero.

$$X^T X W = X^T R$$

3. Closed-Form Solution

Direct matrix solution when $X^T X$ is invertible.

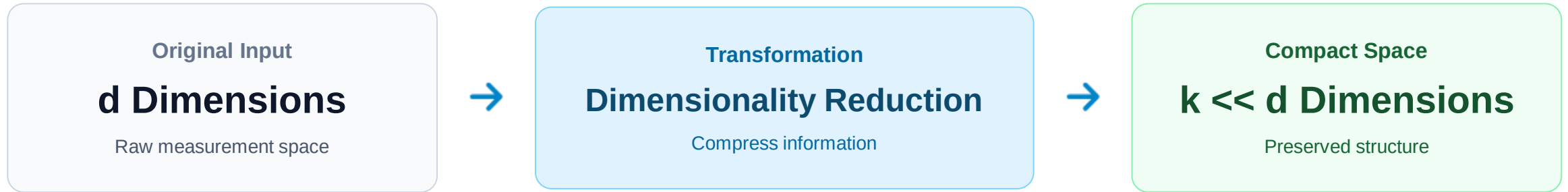
$$W = (X^T X)^{-1} X^T R$$

Summary of Multivariate Methods

Dimension	Discrete Features (Sec 5.7)	Multivariate Regression (Sec 5.8)
Task Goal	Classification from binary/categorical inputs	Simultaneous prediction of k numeric targets
Model Basis	Bernoulli with Naive Bayes independence	Linear projection $R = XW + E$
Estimation	Sample frequency counts $\hat{p}_{ij}$	Normal equations $W = (X^T X)^{-1} X^T R$

Exam Takeaway: Discrete features replace covariance with conditional product probabilities; multivariate regression extends single output r to target matrix R.

What is Dimensionality Reduction?



Textbook Definition: Process of finding a lower-dimensional mapping that preserves relevant data structure while discarding noise and redundant degrees of freedom.

Why Reduce Dimensionality?



Computational Speed

Lowers algorithmic time and storage requirements.



Avoid Overfitting

Fewer parameters prevent variance explosion.



Acquisition Cost

Fewer sensor measurements save money.



Data Visualization

2D and 3D visual projection reveals clusters.

Feature Selection vs Feature Extraction



1. Feature Selection

Selects a subset of k existing features from the original d features.

Key Trait: Preserves original physical interpretation.

Method in syllabus: [Subset Selection \(Sec 6.2\)](#)



2. Feature Extraction

Creates k brand new features by linear or non-linear combination.

Key Trait: Maximizes variance or separability.

Method in syllabus: [PCA \(Sec 6.3\)](#)

The Subset Selection Objective

Target Goal

Find subset $F \subset \{x_1, \dots, x_d\}$ with $|F| = k$ that minimizes generalization error.

Discards uninformative and redundant variables.

Feature Types

Relevant: Carries critical target predictive signal.

Redundant: Strongly correlated with already selected attributes.

Evaluation Function

Textbook uses validation set error $E(F)$ to evaluate candidate subsets.

Prevents training set memorization.

Forward Feature Selection

Algorithm Procedure

1. **Initialize:** Start with empty set $F = \emptyset$.
2. **Evaluate:** Train model with $F \cup \{x_i\}$ for each candidate.
3. **Select:** Add feature x^* that minimizes validation error.
4. **Stopping Rule:** Halt if error reduction drops below threshold.

CLASSROOM ILLUSTRATION

Sample Trace (d = 4 features)

Step 1: Compare $\{x_1\}, \{x_2\}, \{x_3\}, \{x_4\} \rightarrow$ **Pick x_2**

Step 2: Compare $\{x_2, x_1\}, \{x_2, x_3\}, \{x_2, x_4\} \rightarrow$ **Pick x_4**

Step 3: Adding x_1 or x_3 increases error $\rightarrow$ **Stop**

Final Selected Subset: $\{x_2, x_4\}$

Backward Feature Selection

Algorithm Procedure

- 1. **Initialize:** Start with all features $F = \{x_1, \dots, x_d\}$.
- 2. **Evaluate:** Measure error removing one candidate at a time.
- 3. **Remove:** Drop feature that harms validation score least.
- 4. **Stopping Rule:** Halt when further removal causes error spike.

Aspect	Forward Selection	Backward Selection
Start Point	Empty set $\emptyset$	Full set $\{x_1, \dots, x_d\}$
Cost (d large)	Low if target k is small	High (trains full model)
Small N Issue	Works when $N < d$	Fails if $N < d$ (singular matrix)

Limitations of Subset Selection



Combinatorial Explosion

Exhaustive search requires checking $2^d - 1$ subsets.

For $d = 30$, subsets exceed 10^9 .



Greedy Suboptimality

Forward and backward heuristics make locally greedy decisions.

Cannot guarantee globally optimal subset.



Irreversible Choices

Forward search cannot drop previously added features later.

Backward search cannot re-evaluate dropped features.

Principal Components Analysis (PCA)

What is PCA?

An **unsupervised**, linear feature extraction method.

Transforms original correlated attributes into new mutually uncorrelated variables.

Core Objective

Find orthogonal directions that capture maximum data variance.

Higher variance retains higher information content.

Understanding Principal Components

1st PC

Maximum Variance

Direction capturing largest spread of sample points.

2nd PC

Orthogonal Direction

Perpendicular to 1st component with next highest variance.

Ranked

Ordered Variance

Variance decreases systematically across components.

Zero Cov

Uncorrelated

Covariance between distinct components is exactly zero.

Eigenvalues and Eigenvectors in PCA

Eigen-Decomposition of Covariance

Linear projection: $z_1 = w_1^T (x - m)$ with $\|w_1\| = 1$.

Maximizing variance $\text{Var}(z_1) = w_1^T \Sigma w_1$ yields:

$$\Sigma w_i = \lambda_i w_i$$

Proportion of variance explained by k components: $\sum_{i=1}^k \lambda_i / \sum_{j=1}^d \lambda_j$

Symbols Explained

Σ (or S): $d \times d$ sample covariance matrix

w_i (Eigenvector): Direction vector of principal axis

λ_i (Eigenvalue): Magnitude of data variance along w_i

z_i : Score value in transformed reduced space

PCA Step-by-Step: 2D to 1D

Step 1

Center the Data

Subtract sample mean vector m from each observation.

Step 2

Compute Covariance

Calculate 2×2 sample covariance matrix S .

Step 3

Extract Eigenvectors

Sort eigenvalues $\lambda_1 \geq \lambda_2$ and directions w_1, w_2 .

Step 4

Project to 1D

Compute $z_1 = w_1^T(x - m)$, dropping minor axis w_2 .

Subset Selection vs PCA

Criterion	Subset Selection (Sec 6.2)	Principal Components Analysis (Sec 6.3)
Technique Type	Feature Selection	Feature Extraction
Output Attributes	Retains subset of original features	Creates new linear combinations (PCs)
Supervision	Supervised (guided by validation error)	Unsupervised (guided by variance)
Interpretability	Direct physical meaning preserved	Abstract linear combination weights

Key Takeaways & Model Questions

Short Answer Questions (2–4 Marks)

1. Define discrete features in Alpaydin's formulation.
2. Write the Bernoulli naive Bayes probability equation.
3. Differentiate multiple regression from multivariate regression.
4. State the closed-form least-squares estimator W .
5. Distinguish feature selection from feature extraction.
6. Define the first principal component in PCA.

Long Answer Questions (8–10 Marks)

1. Explain parameter estimation for discrete features with Naive Bayes.
2. Derive multivariate regression and explain parameter matrix W .
3. Explain motivations for dimensionality reduction from Section 6.1.
4. Compare forward selection and backward selection with limitations.
5. Explain PCA mathematical formulation using eigen-decomposition.
6. Compare Subset Selection and PCA on five key dimensions.